

Short Concept Questions

- 3.1. Under what circumstances are short hedges and long hedges appropriate?
- 3.2. Give three reasons why the treasurer of a company might choose not to hedge a particular risk.
- 3.3. Explain how basis risk arises in hedging.
- 3.4. What is the formula for the minimum variance hedge ratio when daily settlement is ignored?
- 3.5. How is the formula for the minimum variance hedge ratio changed to take account of daily settlement?
- 3.6. How is the number of contracts used for hedging calculated from the minimum variance hedge ratio?
- 3.7. How can index futures be used to change the beta of a well-diversified portfolio?
- 3.8. How might investors who consider themselves adept at stock picking use index futures?
- 3.9. Explain what the stack and roll hedging strategy involves.
- 3.10. Explain how hedging using futures can lead to cash-flow problems such as those experienced by Metallgesellschaft.

Practice Questions

- 3.11. Explain what is meant by a *perfect hedge*. Does a perfect hedge always lead to a better outcome than an imperfect hedge? Explain your answer.
- 3.12. Under what circumstances does a minimum variance hedge portfolio lead to no hedging at all?
- 3.13. Suppose that the standard deviation of quarterly changes in the prices of a commodity is \$0.65, the standard deviation of quarterly changes in a futures price on the commodity is \$0.81, and the coefficient of correlation between the two changes is 0.8. What is the optimal hedge ratio for a 3-month contract? What does it mean?
- 3.14. A company has a \$20 million portfolio with a beta of 1.2. It would like to use futures contracts on a stock index to hedge its risk. The index futures price is currently standing at 1080, and each contract is for delivery of \$250 times the index. What is the hedge that minimizes risk? What should the company do if it wants to reduce the beta of the portfolio to 0.6?
- 3.15. In the corn futures contract traded on an exchange, the following delivery months are available: March, May, July, September, and December. Which of the available contracts should be used for hedging when the expiration of the hedge is in (a) June, (b) July, and (c) January.
- 3.16. Does a perfect hedge always succeed in locking in the current spot price of an asset for a future transaction? Explain your answer.
- 3.17. Explain why a short hedger's position improves when the basis strengthens unexpectedly and worsens when the basis weakens unexpectedly.

- 3.18. Imagine you are the treasurer of a Japanese company exporting electronic equipment to the United States. Discuss how you would design a foreign exchange hedging strategy and the arguments you would use to sell the strategy to your fellow executives.
- 3.19. Suppose that in Example 3.2 of Section 3.3 the company decides to use a hedge ratio of 0.8. How does the decision affect the way in which the hedge is implemented and the result?
- 3.20. “If the minimum variance hedge ratio is calculated as 1.0, the hedge must be perfect.” Is this statement true? Explain your answer.
- 3.21. “If there is no basis risk, the minimum variance hedge ratio is always 1.0.” Is this statement true? Explain your answer.
- 3.22. “When the futures price of an asset is less than the spot price, long hedges are likely to be particularly attractive.” Explain this statement.
- 3.23. The standard deviation of monthly changes in the spot price of live cattle is (in cents per pound) 1.2. The standard deviation of monthly changes in the futures price of live cattle for the closest contract is 1.4. The correlation between the futures price changes and the spot price changes is 0.7. It is now October 15. A beef producer is committed to purchasing 200,000 pounds of live cattle on November 15. The producer wants to use the December live cattle futures contracts to hedge its risk. Each contract is for the delivery of 40,000 pounds of cattle. What strategy should the beef producer follow?
- 3.24. A corn farmer argues “I do not use futures contracts for hedging. My real risk is not the price of corn. It is that my whole crop gets wiped out by the weather.” Discuss this viewpoint. Should the farmer estimate his or her expected production of corn and hedge to try to lock in a price for expected production?
- 3.25. On July 1, an investor holds 50,000 shares of a certain stock. The market price is \$30 per share. The investor is interested in hedging against movements in the market over the next month and decides to use an index futures contract. The index futures price is currently 1,500 and one contract is for delivery of \$50 times the index. The beta of the stock is 1.3. What strategy should the investor follow? Under what circumstances will it be profitable?
- 3.26. Suppose that in Table 3.5 the company decides to use a hedge ratio of 1.5. How does the decision affect the way the hedge is implemented and the result?
- 3.27. An airline executive has argued: “There is no point in our using oil futures. There is just as much chance that the price of oil in the future will be less than the futures price as there is that it will be greater than this price.” Discuss the executive’s viewpoint.
- 3.28. Suppose that the 1-year gold lease rate is 1.5% and the 1-year risk-free rate is 5.0%. Both rates are compounded annually. Use the discussion in Business Snapshot 3.1 to calculate the maximum 1-year gold forward price Goldman Sachs should quote to the gold-mining company when the spot price is \$1,200.
- 3.29. The expected return on the S&P 500 is 12% and the risk-free rate is 5%. What is the expected return on an investment with a beta of (a) 0.2, (b) 0.5, and (c) 1.4?
- 3.30. It is now June. A company knows that it will sell 5,000 barrels of crude oil in September. It uses the October CME Group futures contract to hedge the price it will receive. Each contract is on 1,000 barrels of “light sweet crude.” What position should it take? What price risks is it still exposed to after taking the position?
- 3.31. Sixty futures contracts are used to hedge an exposure to the price of silver. Each futures contract is on 5,000 ounces of silver. At the time the hedge is closed out, the basis is \$0.20

- per ounce. What is the effect of the basis on the hedger's financial position if (a) the trader is hedging the purchase of silver and (b) the trader is hedging the sale of silver?
- 3.32. A trader owns 55,000 units of a particular asset and decides to hedge the value of her position with futures contracts on another related asset. Each futures contract is on 5,000 units. The spot price of the asset that is owned is \$28 and the standard deviation of the change in this price over the life of the hedge is estimated to be \$0.43. The futures price of the related asset is \$27 and the standard deviation of the change in this over the life of the hedge is \$0.40. The coefficient of correlation between the spot price change and futures price change is 0.95.
- (a) What is the minimum variance hedge ratio?
 - (b) Should the hedger take a long or short futures position?
 - (c) What is the optimal number of futures contracts when adjustments for daily settlement are not considered?
 - (d) How can the daily settlement of futures contracts be taken into account?
- 3.33. A company wishes to hedge its exposure to a new fuel whose price changes have a 0.6 correlation with gasoline futures price changes. The company will lose \$1 million for each 1 cent increase in the price per gallon of the new fuel over the next three months. The new fuel's price changes have a standard deviation that is 50% greater than price changes in gasoline futures prices. If gasoline futures are used to hedge the exposure, what should the hedge ratio be? What is the company's exposure measured in gallons of the new fuel? What position, measured in gallons, should the company take in gasoline futures? How many gasoline futures contracts should be traded? Each contract is on 42,000 gallons.
- 3.34. A portfolio manager has maintained an actively managed portfolio with a beta of 0.2. During the last year, the risk-free rate was 5% and equities performed very badly providing a return of -30% . The portfolio manager produced a return of -10% and claims that in the circumstances it was a good performance. Discuss this claim.
- 3.35. It is July 16. A company has a portfolio of stocks worth \$100 million. The beta of the portfolio is 1.2. The company would like to use the December futures contract on a stock index to change the beta of the portfolio to 0.5 during the period July 16 to November 16. The index futures price is currently 2,000 and each contract is on \$250 times the index.
- (a) What position should the company take?
 - (b) Suppose that the company changes its mind and decides to increase the beta of the portfolio from 1.2 to 1.5. What position in futures contracts should it take?